

Camera Connector / Pinout Comparison

Chronos **HW_CHRONOS_R1** (Panasonic AXT524124) vs Waveshare **OV9281-120** (Raspberry Pi 15-pin FFC)

Key finding

The Waveshare OV9281-120 uses the standard Raspberry Pi 15-pin CSI interface. This connector **does NOT provide an FSIN (frame-sync / external-trigger) pin**, and the module free-runs on its own on-board 24 MHz oscillator. Chronos requires FSIN to capture all 4 cameras in hardware sync (global-shutter, simultaneous). Hardware sync with this module is therefore only possible by soldering a wire to the OV9281 FSIN pad on the Waveshare PCB. A module that breaks FSIN out on the connector (e.g. Arducam with FSIN/STROBE header) avoids this.

Signal comparison

Function	Chronos AXT524124 (pin)	Waveshare RPi 15-pin (pin)	Status
MIPI Clock +/-	4 / 5	9 / 8	Present (compatible)
MIPI Data Lane 0 +/-	7 / 8	3 / 2	Present
MIPI Data Lane 1 +/-	10 / 11	6 / 5	Present
MIPI Data Lane 2 +/-	13 / 14	—	Not on Waveshare (2-lane)
MIPI Data Lane 3 +/-	20 / 23	—	Not on Waveshare (2-lane)
SCCB SCL / SDA	2 / 1 (1.8 V)	13 / 14 (3.3 V)	Present, different voltage
FSIN (hardware sync)	16 (1.8 V)	— (none)	NOT on Waveshare connector
XCLK / MCLK (sensor clock)	19	—	Waveshare self-clocked (24 MHz)
Camera Reset	21 (CAM_RESET)	— (POWER-EN pin 11)	Different
Power	2.8 V (18), 1.8 V (22), 1.2 V (3)	single 3.3 V (15)	Different power scheme
POWER-EN	—	11	Waveshare only
LED-EN	—	12	Waveshare only
GND / AGND	6, 9, 12, 15, 17, 24 (+25-28)	1, 4, 7, 10	Present

Full pinout reference

Chronos AXT524124 (CAM0 = J3)		Waveshare OV9281-120 (RPi 15-pin)	
1	CAM0_SDA (1.8 V)	1	GND
2	CAM0_SCL (1.8 V)	2	MDN0 (D0-)
3	Supply ~1.2 V	3	MDP0 (D0+)
4 / 5	CAM0_CLK_P / _N	4	GND
6	GND	5	MDN1 (D1-)
7 / 8	CAM0_D0_P / _N	6	MDP1 (D1+)
9	GND	7	GND
10 / 11	CAM0_D1_P / _N	8	MCN (CLK-)
12	GND	9	MCP (CLK+)
13 / 14	CAM0_D2_P / _N (unused)	10	GND
15	AGND	11	POWER-EN
16	FSIN0 (1.8 V)	12	LED-EN
17	GND	13	SCL (3.3 V)
18	Supply ~2.8 V	14	SDA (3.3 V)
19	CAM0_MCLK	15	VCC 3.3 V
20 / 23	CAM0_D3_P / _N (unused)	No FSIN / XCLK / reset pins.	
21	CAM_RESET		
22	Supply ~1.8 V		
24 / 25-28	GND / mounting		